



Regression Analysis of Goodreads Book Ratings

Iris Wu

Data 201

Data Overview

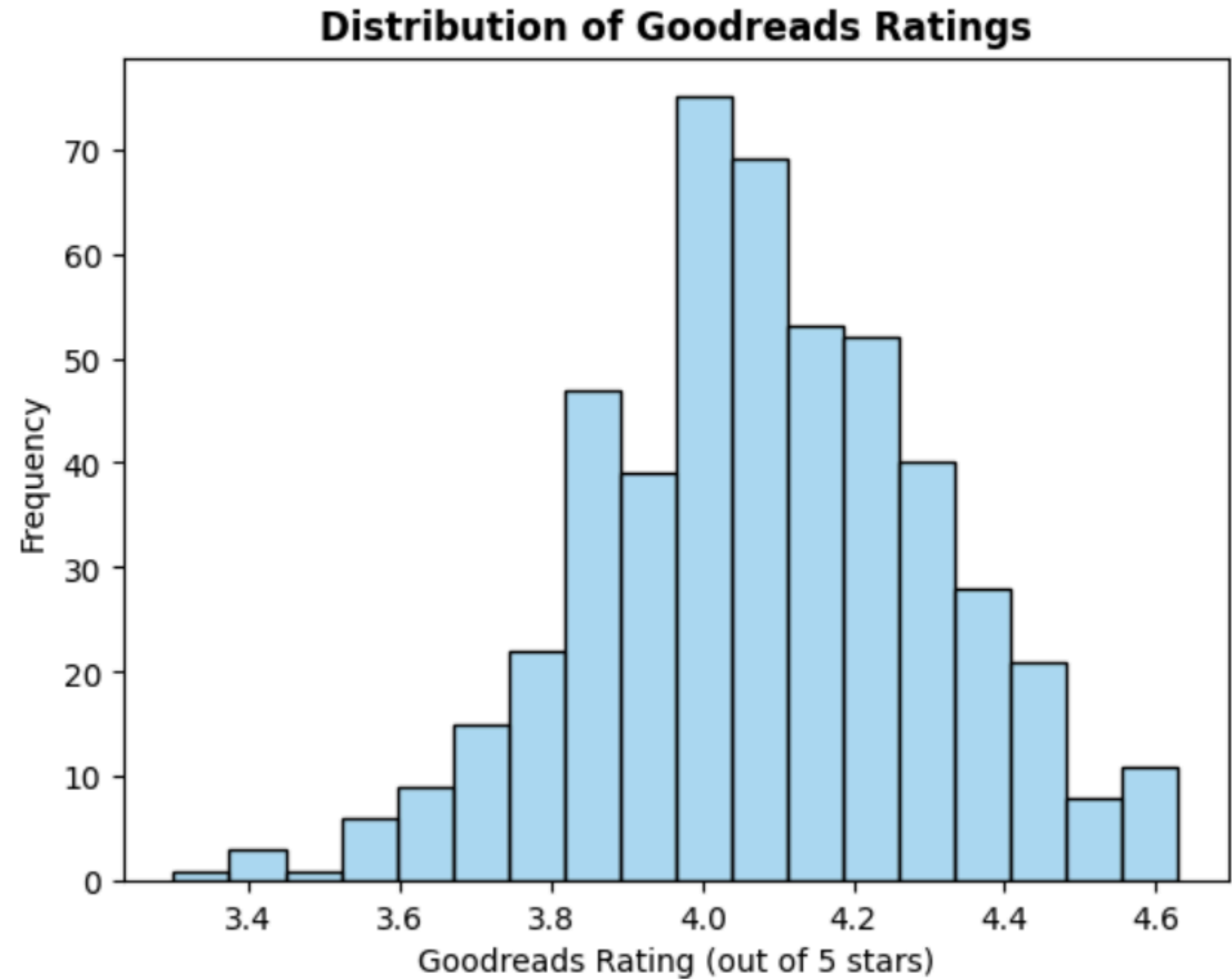
- Best Books Ever dataset: 52,748 records and 25 variables
- Narrowed down to 500 records and 8 variables

Name		Definition	Type
title	Title of book		Categorical
author	Author of book		Categorical
rating	Goodreads rating of book out of 5 stars		Quantitative
pages	Number of pages in book		Quantitative
awards	Literary awards that the book won: converted into a binary column where has_award is 1 and no_award is 0		Categorical
numRatings	Total number of ratings		Quantitative
ratingsByStars	Number of ratings by star: filtered for number of 5-star ratings		Quantitative
bbeVotes	Number of votes received for the Best Books Ever list		Quantitative

Exploratory Plots

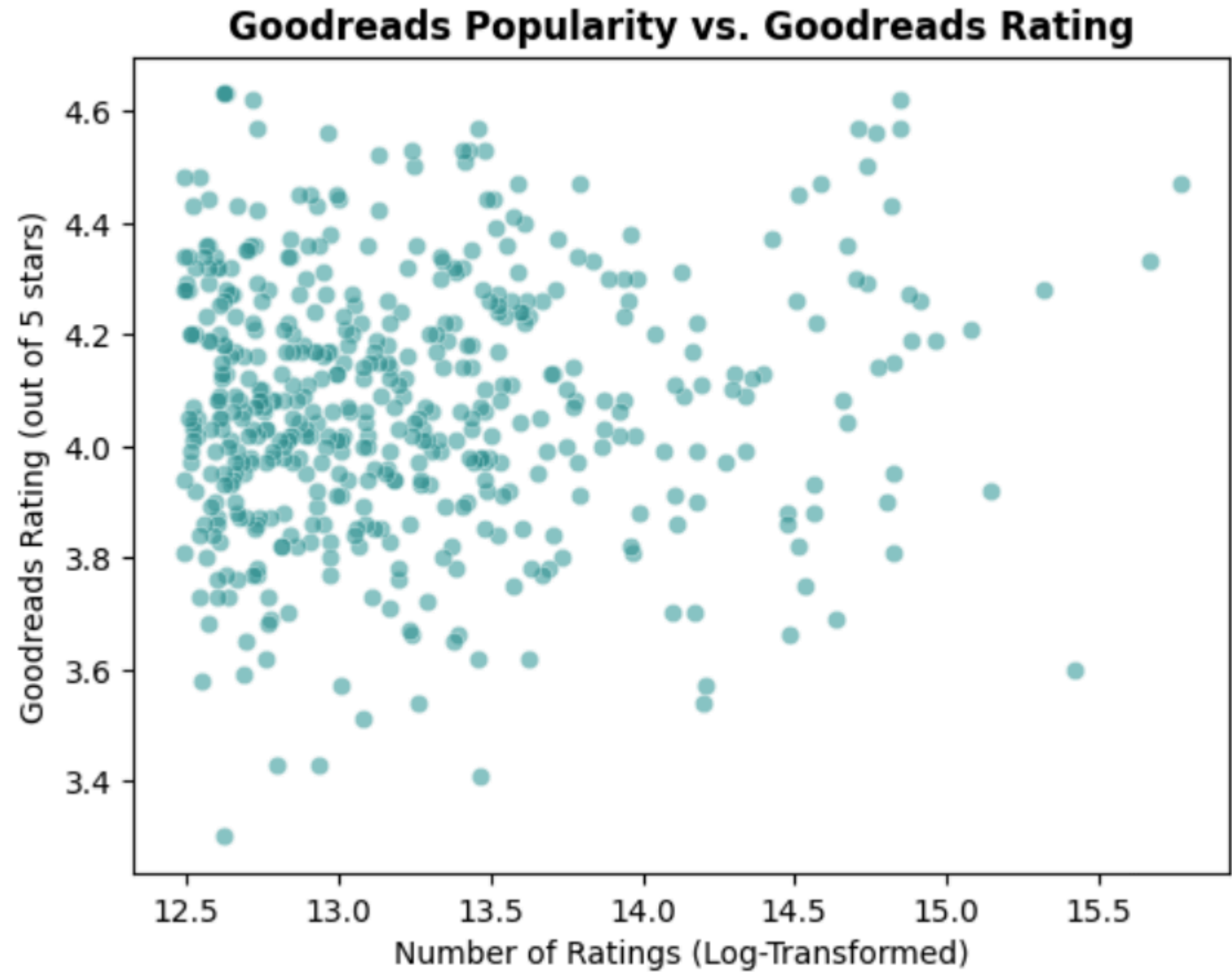


Slightly left-skewed distribution:
most users rate books favorably.



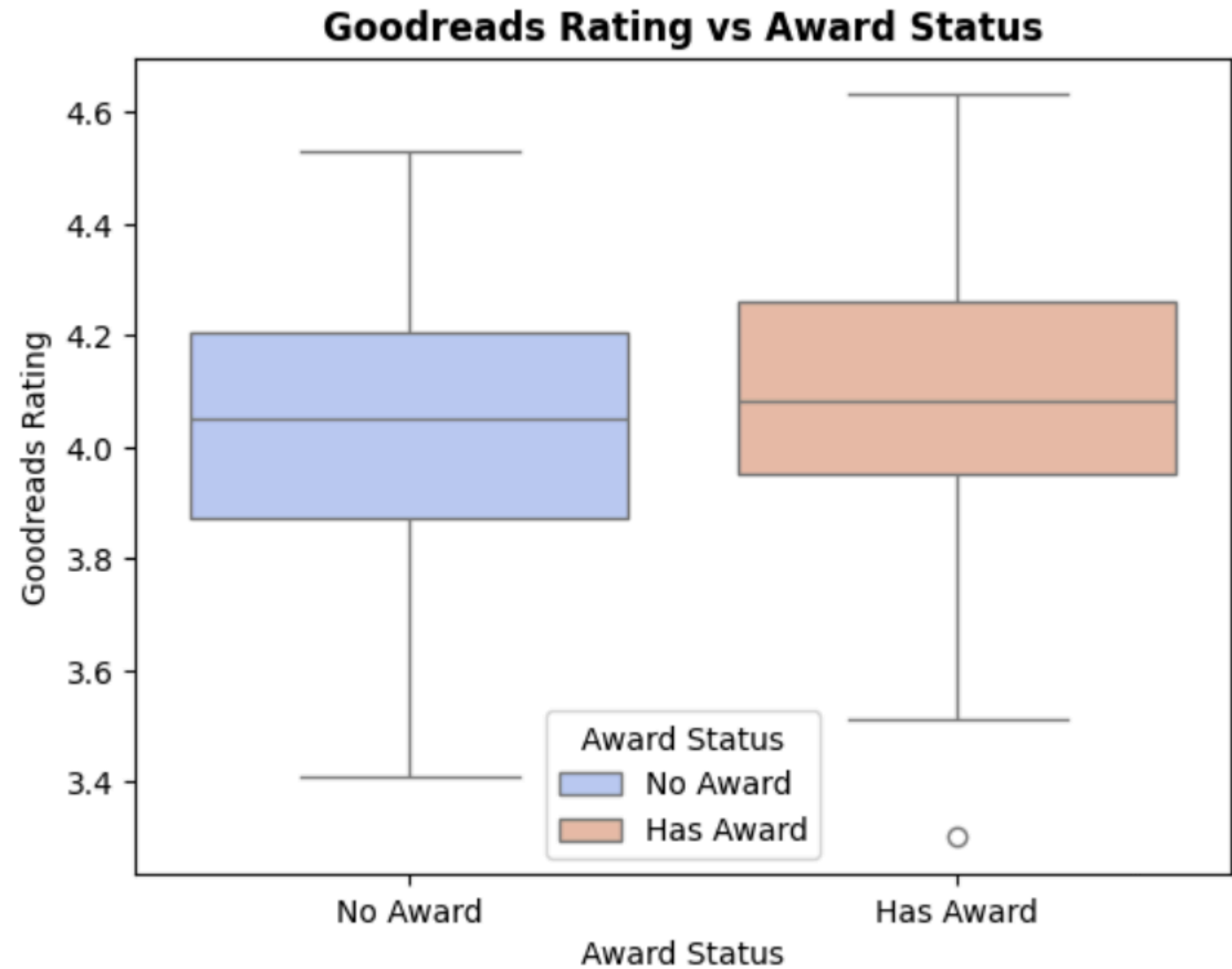
Source: Lorena Casanova Lozano, & Sergio Costa Planells. (2020)

Popularity does not
correlate with the rating of
a book.



Source: Lorena Casanova Lozano, & Sergio Costa Planells. (2020)

Books that won awards have a slightly higher median rating, but there is not a significant difference.



Source: Lorena Casanova Lozano, & Sergio Costa Planells. (2020)

Linear Regression Modeling



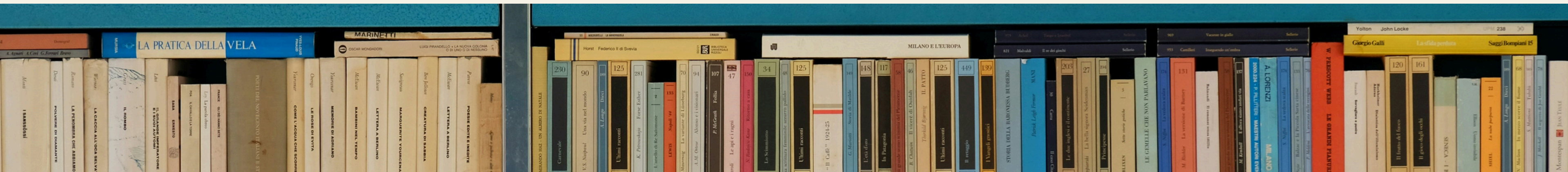
What predicts the Goodreads rating of a book?

Features:

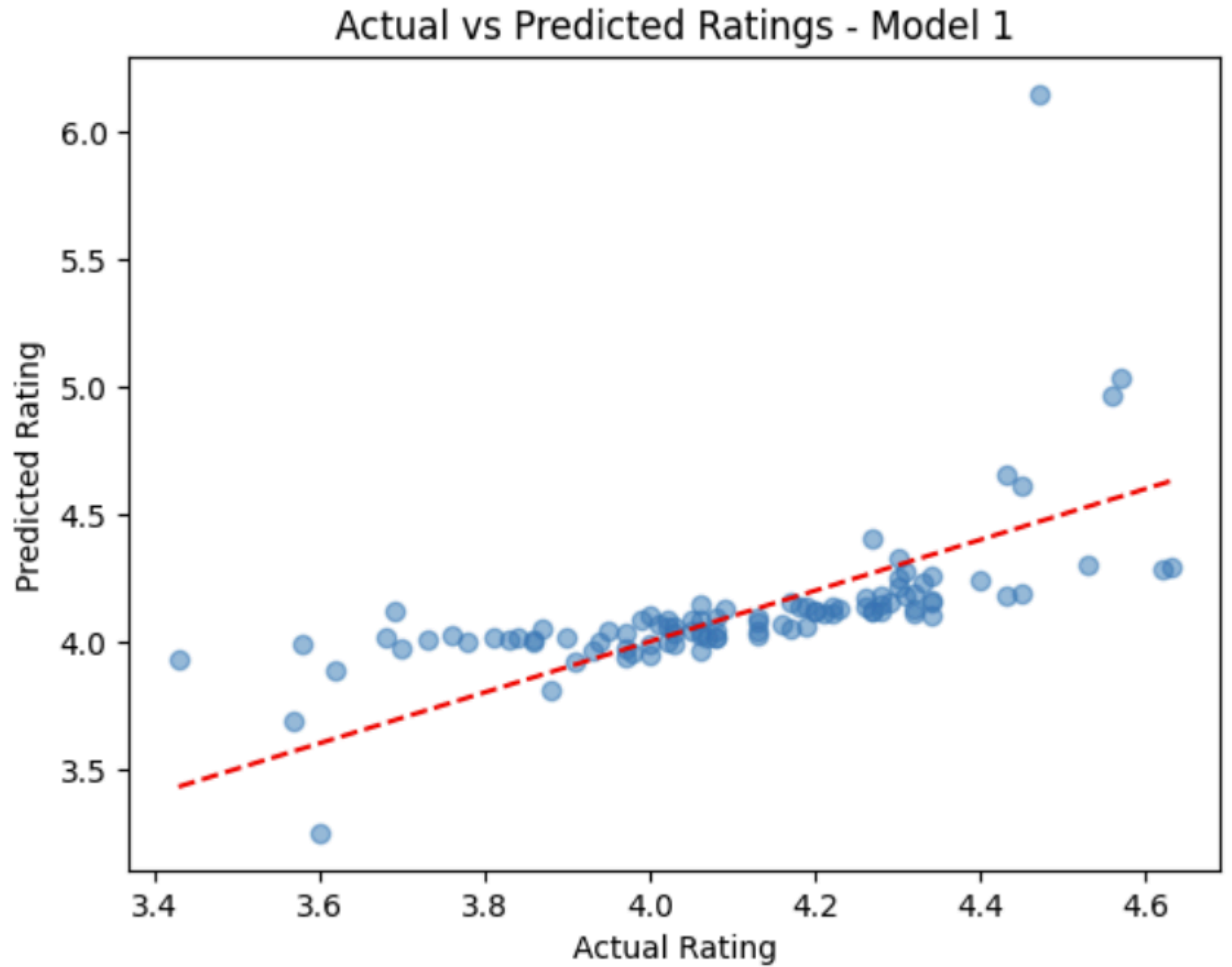
- Number of ratings
- Number of pages
- Number of votes on the Best Books Ever list
- Number of five-star ratings
- Award status

Target:

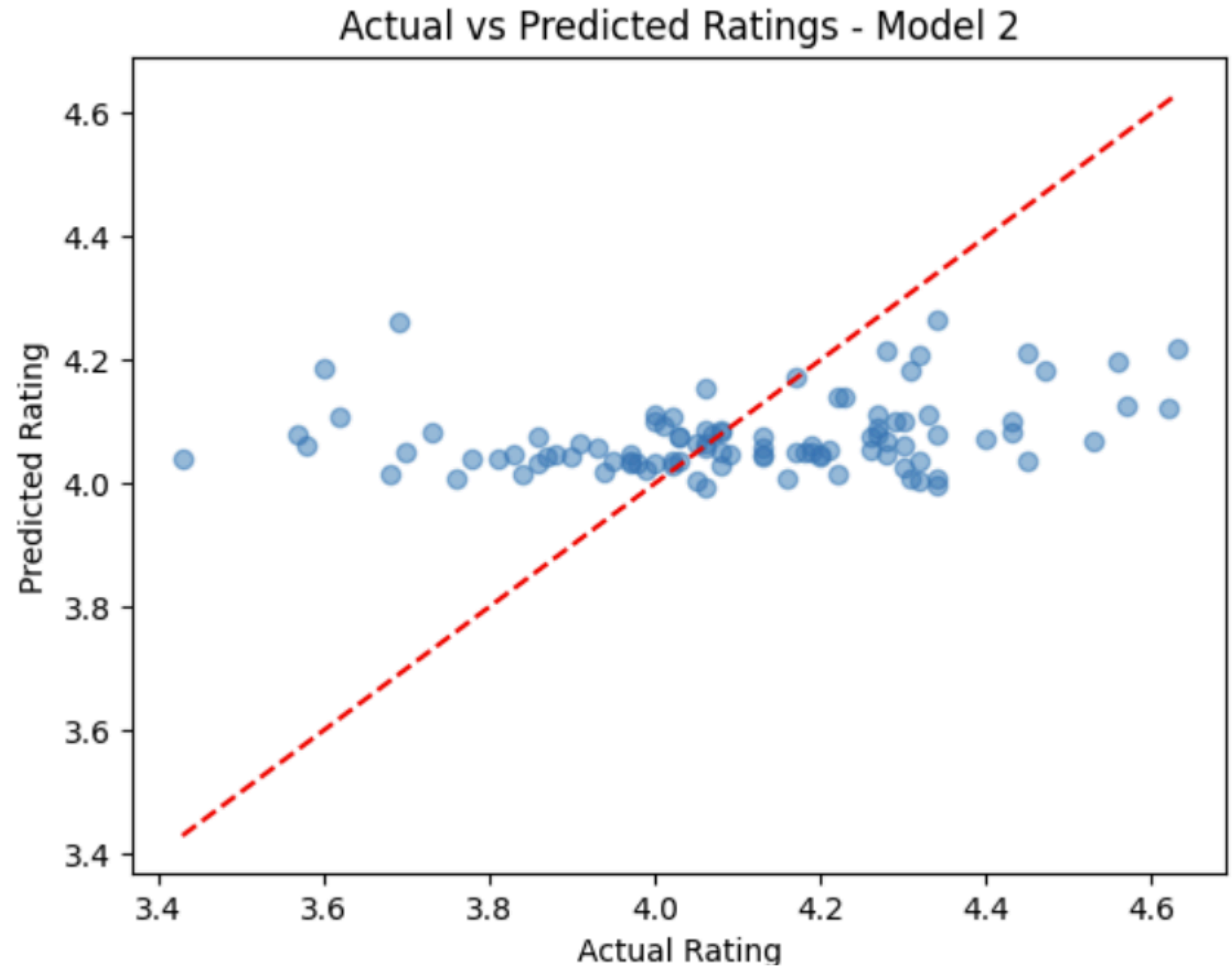
- Rating



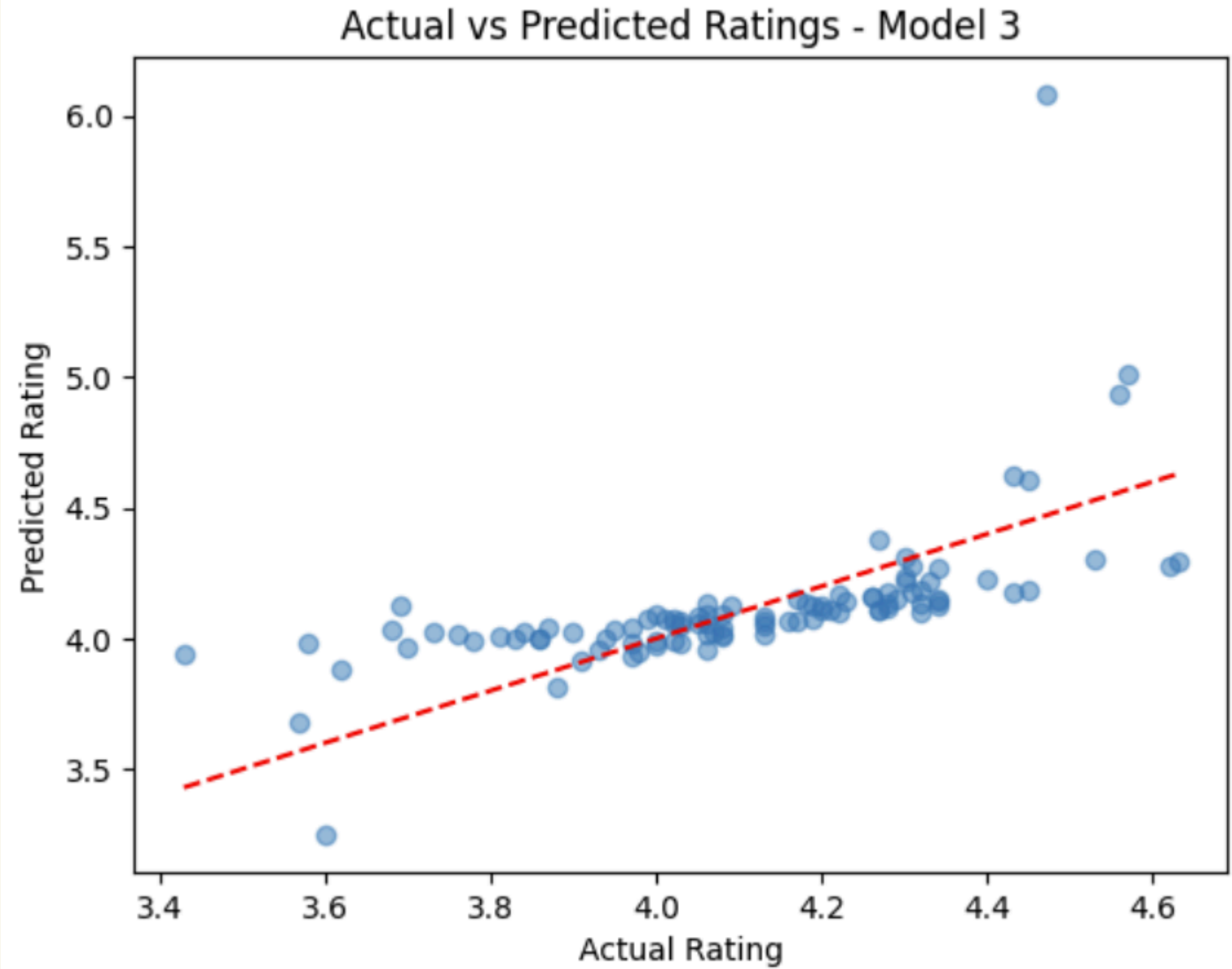
RMSE: 0.2396
Test R-squared: 0.039
Adjusted R-squared: 0.503



RMSE: 0.2379
Test R-squared: 0.052
Adjusted R-squared: 0.058



RMSE: 0.2339
Test R-squared: 0.084
Adjusted R-squared: 0.502



Conclusions



Predicted Goodreads Rating ~ 0.00000154(ratings_5star) + 0.0000962(pages) - 0.000000717(numRatings)

- Individual coefficients are not reliable due to multicollinearity
- Quantitative variables are not good predictors of a book's rating on Goodreads
- Size of the sample may be limiting

